Tissue-resolved metabolic modelling identifies polyol and oxidative C_1 metabolism as the biochemical signature of the *Pleurotus ostreatus* exudophore

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Abstract

Fungal guttation—the exudation of liquid droplets from mycelium—is widely observed and poorly explained. We combined a genome-scale metabolic reconstruction of *Pleurotus ostreatus* with tissue-resolved transcriptomes to ask what distinguishes the metabolism of the exudophore, a discrete exudate-producing structure, from that of vegetative mycelium. Transcriptome-weighted contextualisation produced four tissue-specific models, of which the exudophore model retained the largest reaction network. We show, however, that reaction membership in a context-specific model is a poor guide to tissue-specific biology: of 39 reactions present only in the exudophore model, 28 carried no gene association at all and most of the remainder were expressed as highly in other tissues. Testing expression directly across all 1,899 gene-supported reactions in the network identifies 176 reactions at least twofold elevated in the exudophore, and these converge on a coherent biochemistry: oxidative one-carbon metabolism (alcohol/methanol oxidase and aldehyde dehydrogenase), cyanide detoxification (rhodanese) and oxylipin synthesis (linoleate 8R-dioxygenase), alongside a more modest elevation of polyol metabolism (mannitol and sorbitol dehydrogenase, 1.75-fold). Mannitol is the principal osmolyte of fungal exudates, and alcohol oxidase generates hydrogen peroxide extracellularly; their coordinate elevation offers a mechanistic account of droplet formation and of the oxidative chemistry the droplet may support. We propose specific, measurable predictions for exudate composition.

Keywords: *Pleurotus ostreatus*, genome-scale metabolic model, guttation, mannitol, alcohol oxidase, context-specific model, RIPTiDe

DRAFT — AWAITING AUTHOR INPUT

Requires the morphological description of the exudophore and the sampling protocol ([**AUTHORS**]). Exudophore results rest on two libraries after quality filtering; this is stated throughout rather than in a closing caveat. All values are generated from the analysis repository and verified by `scripts/32_audit_numbers.py`.

1 Introduction

Guttation is among the most visible and least explained fungal behaviours. Droplets form on mycelium across a wide taxonomic range, and their composition—where it has been characterised—includes polyols, sugars, organic acids, proteins and secondary metabolites. Proposed functions range from osmotic regulation and excretion of metabolic by-products to deployment of extracellular enzymes and storage of defensive compounds. These hypotheses make different predictions about the metabolism of the cells that produce the droplets, but testing them requires knowing which cells those are.

Growing *Pleurotus ostreatus* in a mycoponic ceramic-tube system [1] makes discrete mycelial structures accessible for individual sampling, including a distinct exudate-producing structure we term the exudophore. [**AUTHORS: one paragraph describing the exudophore morphologically and stating how it is distinguished from exuding mycelium. This is required for the manuscript to stand.**]

Genome-scale metabolic models offer a way to move from a list of differentially expressed genes to a statement about pathway operation. Constraint-based reconstructions integrate the full stoichiometry of an organism's metabolism, and transcriptome-weighted contextualisation methods extract from them a subnetwork consistent with a particular expression state. Applied to tissues rather than conditions, these methods promise a metabolic account of differentiation. We report both what that approach yielded and a caution about how its output should be interpreted.

2 Results

2.1 Context-specific models differ in size but not in the way that matters

We reconstructed a genome-scale model for *P. ostreatus* BOM_ss5 (5,247 reactions, 2,287 able to carry flux on the defined culture medium; described in the companion resource paper) and contextualised it separately against each tissue transcriptome using RIPTiDe [2], which weights reactions by transcript abundance and requires no expression threshold. All 881 model genes were represented in the expression matrix.

The four models differ in size, and the exudophore retains the largest network: 124 reactions against 109 for exuding mycelium, 100 for fuzzy mycelium and 93 for nodule (Fig. 1a,b). Taken at face value this suggests a metabolically expansive tissue, and 39 reactions appear in the exudophore model and in none of the other three.

That interpretation does not survive checking. Of those 39 reactions, **28 carry no gene association whatsoever**—they are transport and exchange reactions whose presence reflects what the network needs in order to carry flux, not what the tissue transcribes. Of the 11 that do have gene support, most are expressed as highly or more highly in nodule than in the exudophore: succinate:NAD^+ oxidoreductase, pyruvate decarboxylase, fructose 1,6-bisphosphatase and ornithine ketoacid aminotransferase all fall into this category. Only alcohol/methanol oxidase and the mannitol/sorbitol dehydrogenase pair are genuinely exudophore-elevated.

Comparing the distribution of expression ratios for reactions unique to the exudophore model against those shared with other models shows the two are barely distinguishable (Fig. 1c). **Membership of a context-specific model is therefore not, by itself, evidence of tissue-specific metabolism.** We report this because the reaction lists such methods produce are commonly interpreted directly, and because the failure is not visible without going back to the expression the model was built from.

2.2 Direct expression testing across the network identifies 176 elevated reactions

We therefore tested the evidence directly. Each model reaction was mapped through its gene–protein–reaction rule to the genes that could catalyse it, and reaction-level expression was summarised per tissue as the maximum across isoenzymes. Of 1,899 reactions with both gene evidence and measurable expression, **176 are at least twofold higher in the exudophore than in the highest of the other three tissues** (Fig. 2a).

These converge on four functional themes.

Polyol metabolism. D-mannitol:$NADP^+$ 2-oxidoreductase and D-sorbitol 2-dehydrogenase are elevated in the exudophore relative to every other tissue, but at 1.75-fold they fall below the twofold cut applied above. They are included here because they are also among the small number of gene-supported reactions that the contextualised models place in the exudophore and in no other tissue, and because of what mannitol is known to do. Mannitol is the dominant soluble carbohydrate of many basidiomycetes and the principal osmolyte reported in fungal exudate droplets. Its synthesis lowers water potential, which is the most parsimonious physical mechanism for drawing water into a forming droplet.

Oxidative one-carbon metabolism. Alcohol/methanol oxidase shows the largest expression contrast of any enzyme in the curated set, and aldehyde dehydrogenase is elevated from a high baseline. Alcohol oxidase oxidises short-chain alcohols with molecular oxygen and releases hydrogen peroxide; aldehyde dehydrogenase disposes of the resulting aldehyde. The same gene supporting the alcohol oxidase reaction was independently identified as a replicate-supported exudophore marker in the transcriptional analysis, by a method that makes no use of the metabolic network.

Cyanide detoxification. Thiosulfate:cyanide sulfurtransferase (rhodanese) is elevated approximately fourfold. The transcriptional analysis independently recovered cyanide hydratase and cyanate hydratase among the replicate-supported markers. Two enzymatically distinct routes for handling cyanide, recovered by two independent analyses, is more than coincidence deserves.

Oxylipin synthesis. Linoleate 8R-dioxygenase is among the most strongly elevated reactions. Fungal oxylipins act as developmental and quorum signals, and their synthesis in a structure that produces an extracellular liquid phase is suggestive.

2.3 Flux comparison is uninformative on a rich medium

We sampled flux distributions for each contextualised model and compared them. Differences were small and dominated by amino-acid uptake: the mycoponic medium supplies peptone, malt extract and tryptic soy broth, so biosynthetic pathways are bypassed by transport and the models converge on similar solutions. Predicted biomass flux differed by roughly 2% between exudophore and baseline.

This is a limitation of the experimental system rather than of the method, and it is worth stating for others attempting the same comparison: **constraint-based flux contrast between tissues requires a medium restrictive enough to make biosynthesis load-bearing.** On a rich undefined medium the informative signal lies in which reactions are expressed, not in how much flux they carry.

3 Discussion

3.1 A biochemical model of the exudophore

The enzymes elevated in the exudophore compose a coherent scheme. Mannitol synthesis provides the osmotic gradient that draws water into a droplet. Alcohol oxidase, acting on short-chain alcohols, generates hydrogen peroxide in the extracellular space; aldehyde dehydrogenase and formate dehydrogenase dispose of the aldehyde and formate produced, and cytochrome-c peroxidase and peroxisomal catalase—both recovered independently—consume the peroxide. This is a controlled oxidative cycle rather than uncontrolled leakage, and hydrogen peroxide is the obligatory co-substrate of the class II peroxidases that mediate white-rot lignin attack.

The attraction of this scheme is that it makes the droplet functional rather than incidental: a liquid phase in which oxidative chemistry can be conducted at a distance from the hypha, with the osmolyte that creates the phase and the oxidase that charges it upregulated together.

3.2 Predictions

The scheme is testable without further sequencing.

1. Exudate droplets should be mannitol-rich, and mannitol concentration should exceed that of the surrounding mycelium.
2. Droplets should contain hydrogen peroxide, and alcohol oxidase activity should be detectable in droplet fluid.

3. Peroxidase activity in droplets should increase on addition of a suitable alcohol substrate.
4. If oxylipin synthesis is genuinely elevated, 8R-hydroxy fatty acids should be detectable in droplet fluid.

The first two are straightforward assays and would together decide whether the scheme is right.

3.3 Limitations

The exudophore rests on two libraries after quality filtering, and all expression contrasts carry that limitation. The metabolic model is a draft: its biomass objective is coarse and uncurated, mass balance has not been curated, and three reactions were gapfilled on topological rather than genomic evidence. Only 881 of 12,705 genes are represented in the network at all, so any metabolic function without an EC-mapped homologue is invisible here—and lineage-specific secreted proteins, the class most likely to matter in an exudate-producing structure, are exactly the class least likely to be mapped. All enzyme assignments are inferred from homology; none has been tested in this organism. The biochemical scheme above is a hypothesis assembled from converging expression evidence, not a demonstrated pathway.

4 Methods

Culture, sequencing, read processing and the genome-scale reconstruction are described in the companion resource paper; all code is in the accompanying repository.

Contextualisation. The gapfilled BOM_ss5 model was contextualised separately against each tissue with RIPTiDe [2], using DESeq2 [3] size-factor normalised counts for all 12,705 genes with no expression filter, averaged within tissue. Using the `filterByExpr`-filtered matrix instead pruned the models to roughly 90 reactions, because a gene absent from that matrix is indistinguishable from a gene expressed at zero; the unfiltered matrix is the correct input. 500 flux samples were drawn per tissue. Model handling used COBRApy [4]; the reconstruction is built on the ModelSEED biochemistry database [5].

Reaction-level expression. Model reactions were mapped through their GPR rules to locus tags via the `protein_id` attribute of the stock GTF CDS records. Reaction expression per tissue was taken as the maximum across isoenzymes, on the grounds that any one suffices to catalyse the reaction. Reactions were ranked by the ratio of exudophore expression to the highest of the other three tissues; a floor of 5 normalised counts was applied to avoid ratios driven by near-zero denominators.

[**AUTHORS: sampling procedure and exudophore morphology, as above.**]

Figures

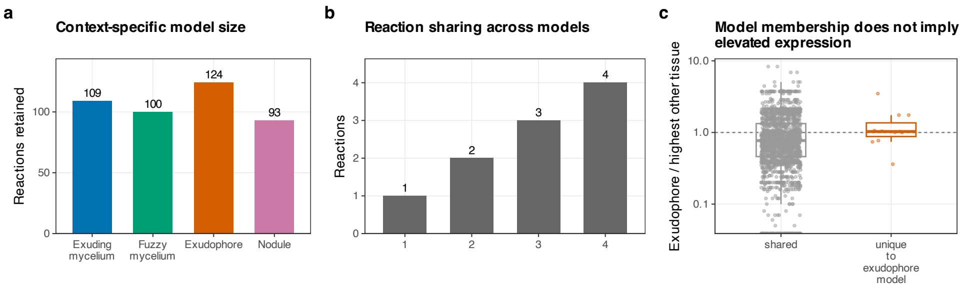

Fig. 1 Context-specific models, and why their reaction lists mislead. a, Reactions retained in each tissue model. **b**, How many tissue models each reaction appears in. **c**, Expression ratio (exudophore over the highest other tissue) for reactions unique to the exudophore model versus reactions shared with other models. The distributions overlap: model membership does not imply elevated expression.

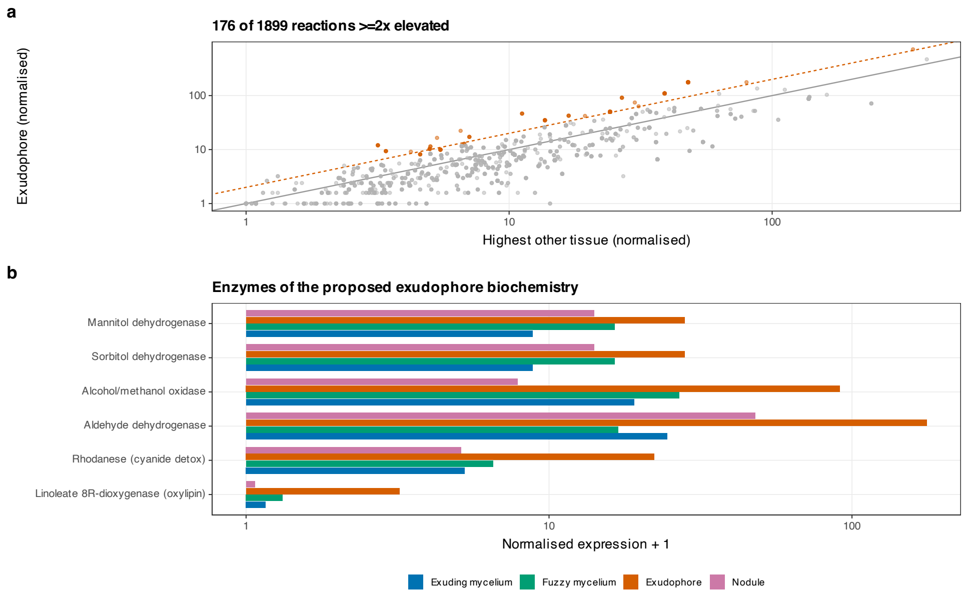

Fig. 2 Reaction-level expression across the metabolic network. a, Exudophore against the highest other tissue for all 1,899 gene-supported reactions; dashed line marks twofold, orange points the 176 reactions exceeding it. **b**, Expression of the enzymes composing the proposed biochemistry. Formate dehydrogenase and cytochrome-c peroxidase are absent from this panel because they carry no gene association in the reconstruction; they enter the argument from the contextualised models and from the transcriptional marker analysis respectively.

Tables

Table 1 Context-specific metabolic models extracted from the 5,247-reaction reconstruction by transcriptome-weighted contextualisation.

Tissue	Libraries	Reactions	Metabolites	Genes
Exuding mycelium	3	109	122	77
Fuzzy mycelium	4	100	108	53
Exudophore	2	124	147	76
Nodule	3	93	109	53

All four were built from the same base model and the same 881 mapped genes.

Table 2 Enzymes composing the proposed exudophore biochemistry. Values are DESeq2-normalised expression averaged within tissue; ratio is exudophore over the highest other tissue.

Enzyme	EC	Reaction	Exudo.	Exuding	Fuzzy	Nodule	Ratio
Alcohol/methanol oxidase	1.1.3.13	rxn00432	90.26	18.07	25.84	6.88	3.49
Aldehyde dehydrogenase	1.2.1.3	rxn00506	175.58	23.5	15.92	46.93	3.74
Mannitol 2-dehydrogenase	1.1.1.138	rxn20173	27.0	7.81	15.45	13.07	1.75
Sorbitol 2-dehydrogenase	1.1.1.14	rxn24346	27.0	7.81	15.45	13.07	1.75
Rhodanese	2.8.1.1	rxn01416	21.26	4.27	5.54	4.13	3.84
Linoleate 8R-dioxygenase	1.13.11.-	rxn04867	2.21	0.16	0.32	0.07	6.9

Formate dehydrogenase and cytochrome-c peroxidase are omitted: they carry no gene association in the reconstruction and so have no reaction-level expression.

Data availability. https://github.com/dr-richard-barker/Myco_tissue_RNAseq. Models in SBML, contextualised models, flux samples and all comparison tables are included.

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Competing interests. [AUTHORS: declaration.]

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